

R Practice Project 4 - TMDb Movie Dataset Analysis

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ALY6000 Introduction to Analytics

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This project focused on the TMDb (Movie Database) set which was a comprehensive collection of over 1,000,000 movies - the information includes about movies involved titles, runtime, release dates, movie genres, and revenue generated. More categories were created such as profits, return on investment and the ability to look at the variables in different decade categories.

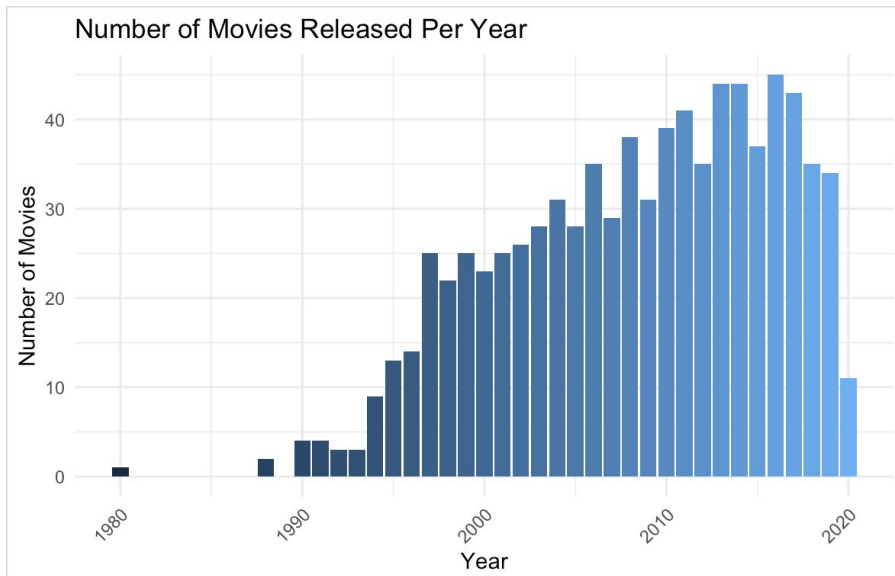
Reviewing the Data Dictionary and Cleaning the Dataset

	decade	total_profit	avg_profit	total_revenue	total_budget	movie_count
1	1980s	522278275	174092758	709278275	187000000	3
2	1990s	22388359867	183511146	32126359867	9738000000	122
3	2000s	67050266582	228062131	97414257982	30363991400	294
4	2010s	137413862025	346130635	188176710083	50762848058	397
5	2020s	1370764229	124614930	2804764229	1434000000	11

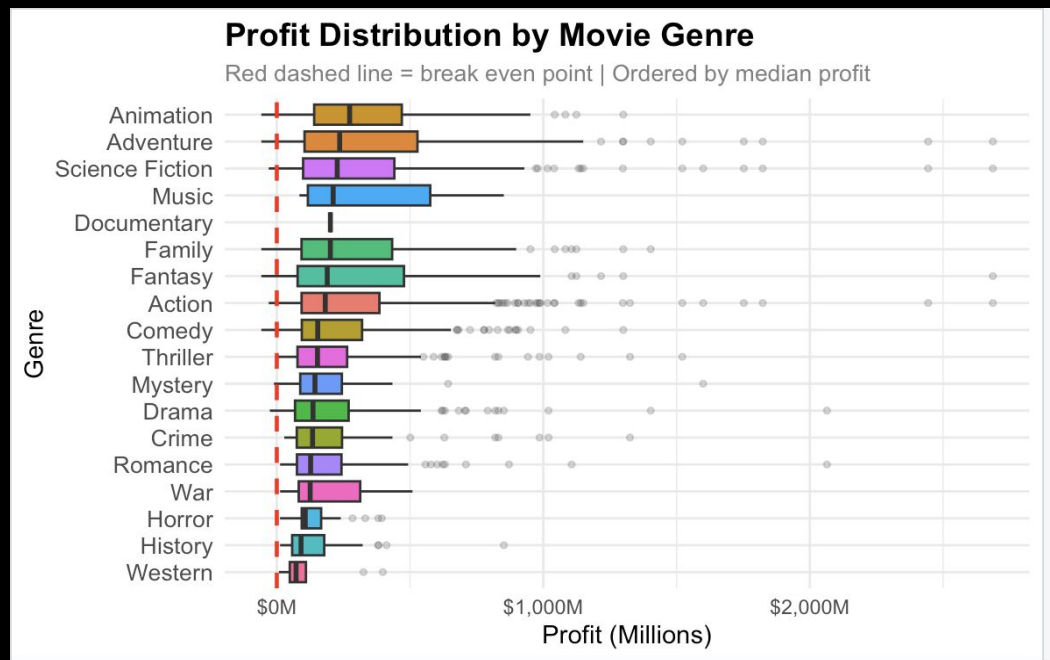
	runtime_category	avg_roi	median_roi	avg_roi_per_minute	movie_count
1	Epic (180+ min),	441.19	335.00	441.19	8
2	Very Long (150–180 min)	303.31	216.36	303.31	64
3	Long (120–150 min)	271.12	211.73	271.12	306
4	Standard (90–120 min)	220.04	168.18	220.04	449

To set up this project, the TMDb movie dataset was imported from Kaggle and the values were cleaned. Specific variables including “homepage”, “backdrop path”, “imdb id”, “adult”, “original title”, “overview”, “poster path”, “tagline”, and “keywords” were removed for analysis purposes. The data was then filtered out to include only movies above 1.5 hours and under 200 minutes - additionally, only movies that were released between 1980 and 2020 were considered for this analysis. Lastly, only movies that generated over 100,000,000 of revenue and had a 50,000,000 budget were considered for this project. The main objectives of this analysis was looking at the movie count over the years and decades as well as the return on investment for specific run time of movies. The data was also analyzed by comparing the growth in budget, revenue, and profit across all decades from 1980 to 2020. The last thing being focused on after cleaning the dataset was looking at the profit distribution and ranges of that by specific movie genre.

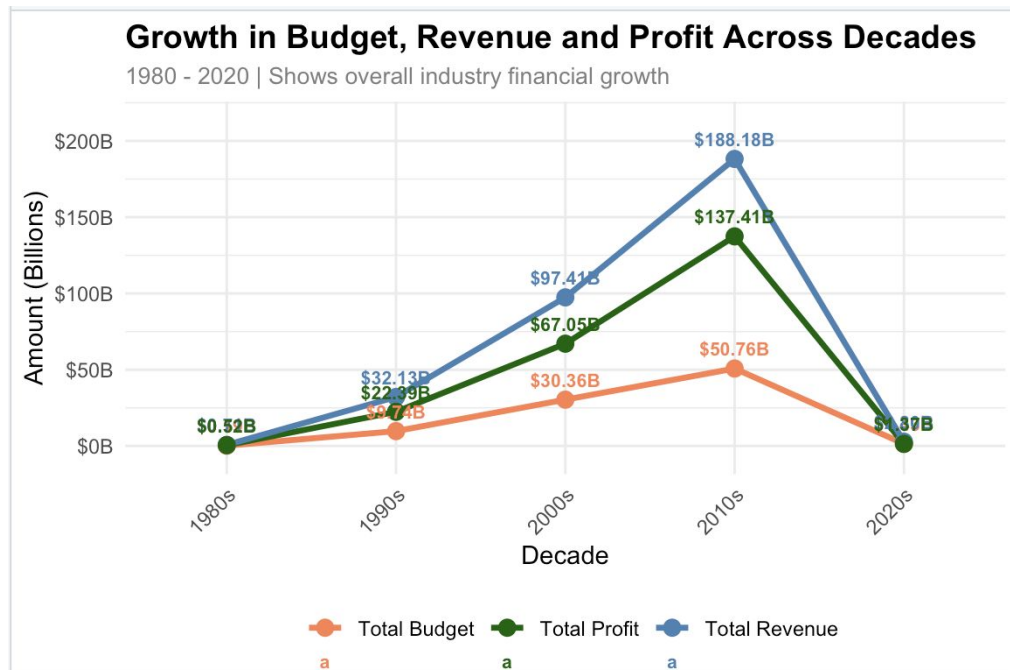
Histogram: Number of Movies Released Per Year



A histogram showing the number of movies released per year highlights the peak era of cinema in the late 2010's where almost 40 movies per year were meeting the criteria to be considered in this dataset and therefore generating over \$100,000,000 of revenue. The 1980's and 1990's highlighted the start of large scale movie production involving this much revenue and budget, likely due to technological advancement and entertainment consumer demand. As movies moved to a larger scale environment in the 2010's, many individuals funding movies were providing lots of money or the movie budget in anticipation of a larger-scale production in revenue generated.



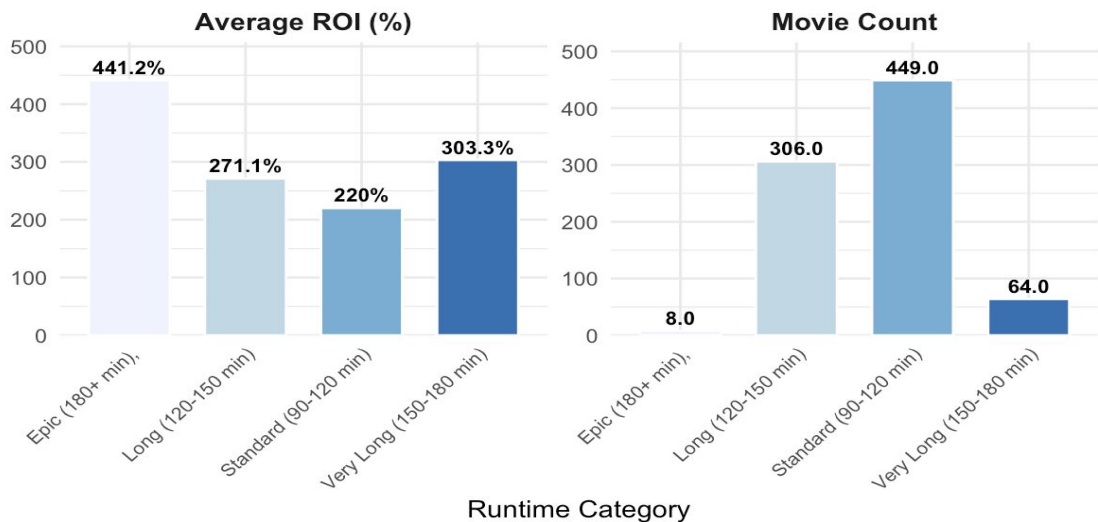
A box plot displays the 18 main movie genres as well as their descriptive statistics which include the mean, median, first/third quartile, and any outliers from the makeup of the plot. Based on the profit distribution in the graph, it shows that Adventure, Action, Fantasy, and Family movies have the widest range between their first and third quartile likely due to the large amounts of production in these genres specifically since they are popular with movie watchers. Furthermore, it is shown that many of these movie genres have at least one outlier in this dataset that made over \$2,000M dollars in profits. It should also be noted that profit was calculated in R Studio using the formula revenue generated minus the budget spent on creating the movie. Lastly, there are a few outlier genres with such a tight range that it is worth pointing out - History, Western, Horror, and Documentary style movies all have a profit that ranges from around approximately 50M - 100M - this is probably due to a lack of demand but when a revenue-generating movie in this genre is produced, it does receive attention from specific fans of that type of movie.



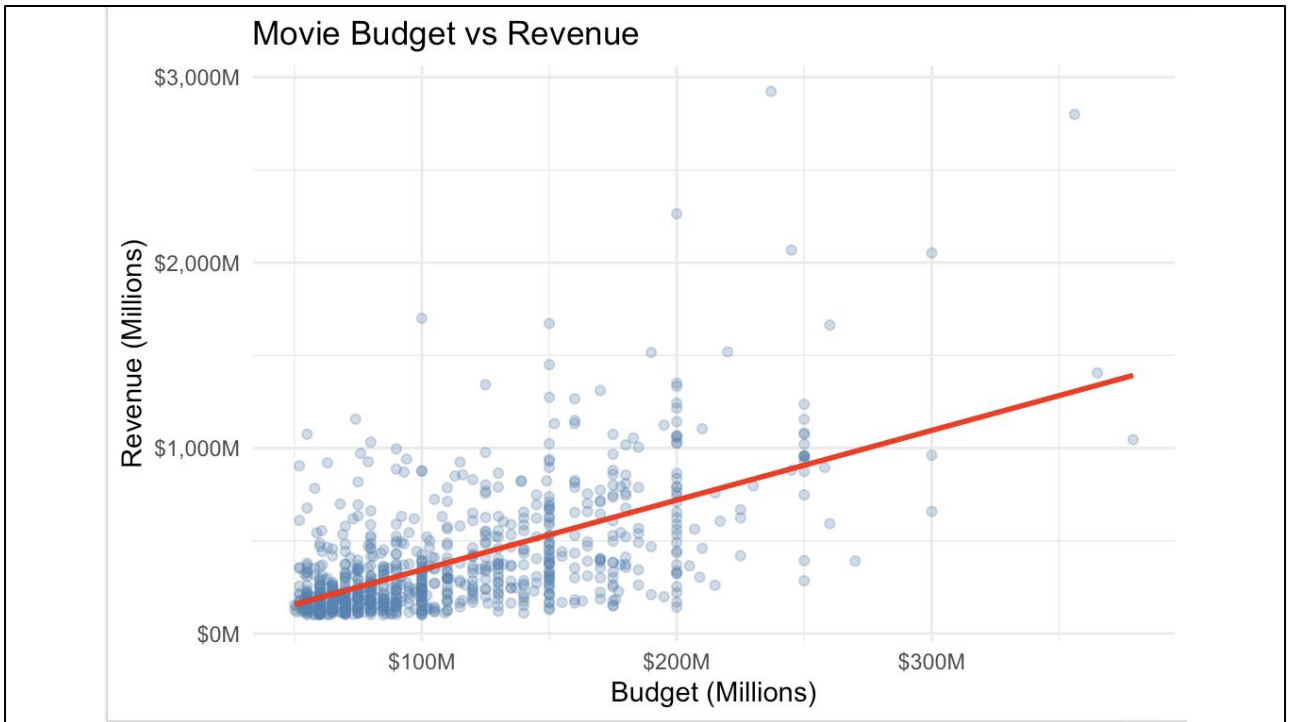
The lineplot shows an advancement over the four decades being analyzed while comparing the total budget, total profit, and total revenue generated for movies produced. In the 1980s movies all three of these statistics were around half a billion dollars - 30 years later in the 2010s, total revenue for movies was sitting at 188 billion dollars, total profit at 137 billion dollars, and the total budget sitting at only 50 billion dollars. This is significant because it means the profit and ultimately the return on investing in producing movies has paid off as consumers are spending more money on the movie industry overall from the 1980s to the 2020s. There is a very clear visual representation of the differences in these three variables and their values by decade with this line plot

Movie Count and Average ROI by Runtime Category

Side by side comparison of volume and return



The histogram shows both the number of movies involved in the dataset per decade, as well as the average return on investment. All of these are categorized by their runtime length which is put into 4 different categories based on differing 30 minute increments. It is very clear that very few movies (only 8) met the criteria for this dataset that were over 180 minutes whereas 449 movies met the criteria in the standard movie category of 90 to 120 minutes. However, it is very interesting to see that the average return on investment for these movies was a lot higher for the longer ones and when there were less overall movies involved. This could be due to the movies that are either very long or epic long being very hyped up and appearing in theaters for a while as a mainstream movie (which is why it generated a higher ROI)



The basic line graph shows the relationship between a movie budget and how much revenue is generated. It is trying to answer the question of if a higher budget generally means that more revenue is generated for the movie. From this graph, it seems as if the relationship is pretty linear with a few clear outliers. This means that for the most part, if a movie has a higher funded budget it is likely generating more revenue. The outliers could be for a number of external factors such as specific actors/actresses being in certain movies, outstanding production, or simply a general consensus from the public that the movie is worth their time/money.

Part III - Communicating what has been learned about the data and follow-up questions

1. When accounting for inflation or other economic conditions, which decade had the highest return on investment?
2. How does the gap between budget, revenue, and ultimately profit change with different genres involved in this dataset over time (decades or years)?
3. Is there an optimal runtime length that consistently produces both movies with enough revenue and profits generated as well as a high enough ROI?
4. How has the dominance of certain genres like action or adventure shifted over time such as years or decades?

What was learned from the data was that movie production started to feature higher budgets, revenues, and ultimately profits starting after the 1980s. This coincided with the number of movies being produced that generated over \$100,000,000 of revenue also starting right after the 1980s. Furthermore the return of interest in the movies that were longer (150 min+) was much higher with a 303% average ROI compared to 271% ROI in any movies under 150 minutes. However, more movies were considered for this dataset inside the 90 min - 150 min range, meaning that these movies may produce more revenue compared to a typical movie that is outside this range. Lastly, the Action, Adventure, and Fantasy movie genres tend to produce higher profits but also have a much wider range of their profit distributions - this is likely due to the high demand and overall larger volume of movies in these genres.

References

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